Nexus Bench: v0 → v1 Verdict

Council aggregation upgrade, measured — 2026-07-16 · branch `nexus/bench-v1` · raw data: `bench/results/`, traces in the Nexus run store

Executive verdict

The v1 aggregator upgrade is **proven**. Changing how the council combines member answers — from freeform synthesis to **vote** (majority over members' answers, winner returned verbatim) — gained **+13.4 points on AIME 2025 (83.3% → 96.7%)** and still edged ahead on MMLU-Pro (94.0% → 94.8%, the best score in either run). Same five member models, same questions, same grading; only the combination policy changed. Separately, fixing two infrastructure defects the benchmark exposed revealed that much of v0's solo-model math ranking was an artifact: Claude and GPT-5.6 Sol were being killed by a hidden 180-second CLI cap, not failing the math.

What changed between the runs

Area	v0	v1
Aggregation	Synthesize only: aggregator reads all members, authors a brand-new answer	vote | verify | synthesize; vote and verify can only SELECT a member's answer verbatim
Mode choice	Manual / always synthesize	Domain classifier + skill matrix picks: math→vote, coding→verify, open-ended→synthesize
Model routing	Hardcoded per-provider strength numbers	Measured per-domain accuracy (skill-matrix.json), 20-pt noise bands, static strengths as fallback
Subscription CLIs	execFileSync — froze the whole server per call; hidden 180s kill cap	Async spawn; NEXUS_CLI_TIMEOUT_MS budget (20 min for math); server stays responsive
SSE streams	No keepalive; clients die at 300s idle	Heartbeat comments every 25s

AIME 2025 — where synthesis was broken (30 problems, exact-answer)

Lane	v0	v1	Δ	Note
nexus-council-vote	**—**	**96.7%**	**+13.4 vs synthesis**	**matches v0 post-hoc prediction exactly**
nexus-council-verify	—	93.3%	+10.0 vs synthesis	auditor can overrule a correct member
nexus-council (synthesize)	83.3%	—		invented compromise answers (see autopsy)
gpt-5.6-sol (solo)	80.0%	100%	+20.0	v0 losses were infra timeouts
chatgpt-sub (solo)	90.0%	96.7%	+6.7	
claude-max (solo)	66.7%	90.0%*	+23.3	*27/30; 3 problems exceed a single CLI turn (23–47 min then internal cap) — 100% of attempted
gemini-flash (solo)	90.0%	not rerun		v0's winner; fast enough to dodge the old caps
deepseek / mistral	53.3%	not rerun		genuine capability ceiling, not timeouts

MMLU-Pro — where synthesis was already good (140 Qs, 14 domains, clean scores)

Lane	v0	v1	Note
nexus-council-vote	**—**	**94.8%**	**best result in either run**
nexus-council (synthesize)	94.0%	—	tied best-solo in v0; synthesis earns its keep here
nexus-council-verify	—	93.2%	
best solo (gemini-flash)	94.0%	—	
claude-max / gpt-5.6-sol	92.5%	—	
ollama-local 8B	48.5%	—	but 90% on health — see skill matrix

HumanEval (v0 only): saturated at the frontier — council/Claude/GPT 99.4%, Gemini 100%; local 8B 57.3%. Needs a harder code suite in v2.

Failure autopsy: why synthesis loses gradable tasks

Compromise hallucination (AIME problem 9, key 81): Gemini solo answered 81 correctly; the v0 council synthesized **77 — a number no member proposed**. The synthesis prompt asks the aggregator to "reconcile contradictions," which is correct behavior for essays and wrong for arithmetic. In v1's trace on the same problem, four members said 81, Mistral dissented with 57, and vote returned 81 (4/5) — the dissent is recorded, not blended in.

Reasoned overrule (mmlupro-11294, live demo for Boris): five members gave five different answers; the synthesis aggregator produced a genuinely rigorous reconciliation — isolated the single point of disagreement (compressibility factor Z), re-derived it via two equations of state, critiqued each member by name — and still landed on E while the key says J (the answer ChatGPT gave). Legible reasoning, single point of failure. Counting is immune to eloquent wrongness; that is the design argument for vote.

The remaining hard tail: AIME problem 13 has never been solved by anything in either run — vote can't rescue a council whose members are all wrong. Aggregation multiplies member competence; it doesn't replace it.

Skill matrix — the routing table Umbruh now uses (per-domain, measured)

Domain signal	Finding	Routing consequence
math	gemini-flash 90%+, never times out; mistral/deepseek ~53–65%	math → gemini or vote-council; never blend-synthesize
law	deepseek 30% vs field ~80%	deepseek is a bargain everywhere except law
health	local 8B holds 90%	health can stay on-device ($0, private)
physics/econ/business	everyone ~100%	route by cost/latency only (deepseek at 1.9s)
coding	frontier saturated (99–100%)	verify-mode council; needs harder v2 suite to rank

Caveat: MMLU domains have n≈10 each — gaps under 20 points are treated as noise by the router; the big gaps (law, math) are the signal.

Methodology & verifiability

All lanes ran through the live Nexus server API (the harness contains no model-calling code). Questions from canonical public datasets (TIGER-Lab MMLU-Pro parquet, math-ai AIME 2025, openai HumanEval); grading is mechanical (exact letter / exact integer / unit-test execution). Identical prompts and extraction across lanes. Every (question, lane) result is one timestamped JSONL line in bench/results/; every council run persists a full member-by-member trace (Run Inspector / /api/runs). Items where ≥5 lanes agreed on the same non-key answer are flagged disputed and reported both included and excluded. Numbers are within-harness comparisons — not comparable to published leaderboard figures; training-data contamination is unknown and inflates all lanes equally.

Recommended next steps

1. Merge nexus/bench-v1 (after nexus/generaluse-sync) and rebuild the packaged app — it still ships the sync-CLI freeze and the old aggregator. 2. Parallelize council members server-side (sequential members are the latency ceiling on hard problems). 3. v2 suites: harder code (LiveCodeBench/SWE-bench), GPQA, IFEval, multimodal for the image-gen routing question, agentic tasks — where a director/worker system should shine most. 4. Marketing claim that survives scrutiny: *"the tuned council matches or beats every model it contains on knowledge and code, is within one problem of the best member on math, and is top-tier on every suite without knowing in advance which model to pick — with member-by-member receipts."*